

-Business Problem Statement

The COVID-19 pandemic significantly impacted public transportation in New York City. The goal of this analysis is to understand ridership trends across subways, buses, railroads, and other transit services, and evaluate recovery rates post-pandemic. Insights will help MTA optimize operations, plan for future disruptions, and improve commuter services.

-Business Processes Overview

Data Collection: Daily ridership from MetroCard/OMNY swipes, ticket sales, and traffic counters.

Data Management: Data is validated, stored, and regularly updated.

Reporting: Key ridership metrics are reported to management for operational decisions.

Decision Making: Analysis informs resource allocation, scheduling, and planning strategies.

-Stakeholders

Stakeholder	Role / Interest
MTA Operations Team	Optimize service schedules and resources
NYC Government	Public transportation planning and policies
Commuters	Reliable and efficient transit service
Data Analysts / Business Intelligence Team	Analyze ridership trends and recovery rates

-Metrics & KPIs

Metrics:

Total Daily Ridership (Subway, Bus, Rail)

% of Pre-Pandemic Ridership

Access-A-Ride Trips

Bridge and Tunnel Traffic

KPIs:

Highest Ridership Day

Lowest Ridership Day

Average Ridership

Recovery Rate (% of Pre-Pandemic)

-Analytical Questions

How did COVID-19 impact ridership across transport modes?

- Which transportation mode recovered fastest?
- What are the peak and lowest ridership days?
- How do weekdays compare to weekends in ridership patterns?
- Are there seasonal trends or anomalies in ridership?

- Data Sources

Source	Description	File Name
MTA Daily Ridership Data	Daily ridership & traffic for all major MTA agencies, including % vs pre-pandemic day	https://drive.google.com/file/d/1MAGZ1etB7hD9dq8uL26MReKCbiKm0ot-/view?usp=drive_link
Data Dictionary	Field definitions, descriptions, and caveats	https://drive.google.com/file/d/1XJBXens-RKCoF32zJKguAmVTvT-AGZb/view?usp=drive_link
Overview Document	Background, methodology, statistical notes, release notes	MTA_DailyRidershipData_Overview.pdf

-Data Dictionary

Column	Description	Type	Notes
Date	Date of record	Date	Format YYYY-MM-DD
Subways: Total Estimated Ridership	Total subway passengers	Numeric	Daily total
Subways: % of Comparable Pre-Pandemic Day	Subway ridership vs pre-COVID	Numeric (%)	Comparison with 2019 same day type
Buses: Total Estimated Ridership	Total bus passengers	Numeric	Daily total
LIRR Ridership	Long Island Rail Road passengers	Numeric	Daily total
Metro-North Ridership	Metro-North passengers	Numeric	Daily total
Access-A-Ride Trips	Trips for eligible disabled customers	Numeric	Scheduled trips
Bridges and Tunnels Traffic	Vehicle traffic	Numeric	Daily total

Staten Island Railway
Ridership

Staten Island rail
passengers

Numeric

Daily total

-Assumptions & Constraints

Assumptions:

- Ridership data accurately reflects daily passenger counts.
- % of Pre-Pandemic uses 2019 averages for comparison.
- Data is representative of normal commuting patterns.

Constraints:

- Some missing data or estimation errors may exist.
- Data does not include rider demographics.
- Only NYC public transit is included; private or alternative transport is excluded.